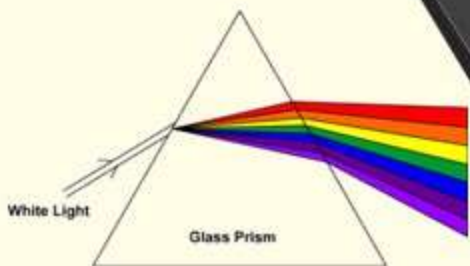


PHYSICS

HUMAN EYE AND THE EYE DEFECTS

MODULE - 8



WHAT DO THESE PICTURES SUGGEST ?



Human eye



Reading glasses



Telescopic glasses



Camera lens

These all are made up with spherical lens.

THE HUMAN EYE

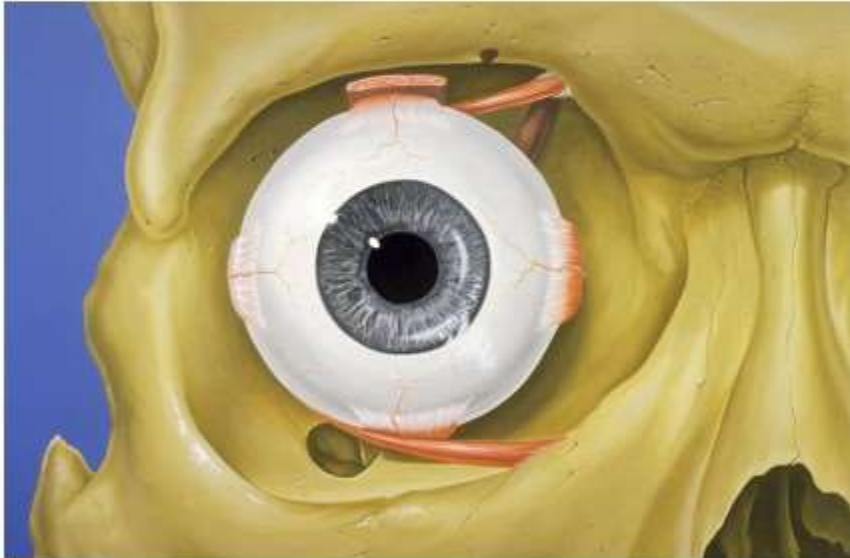


- The Human Eye is a natural optical instrument.
- This is used to see the objects by human beings.
- It is like a camera which has lens and a screen system.



THE HUMAN EYE - ANATOMY

ORBIT



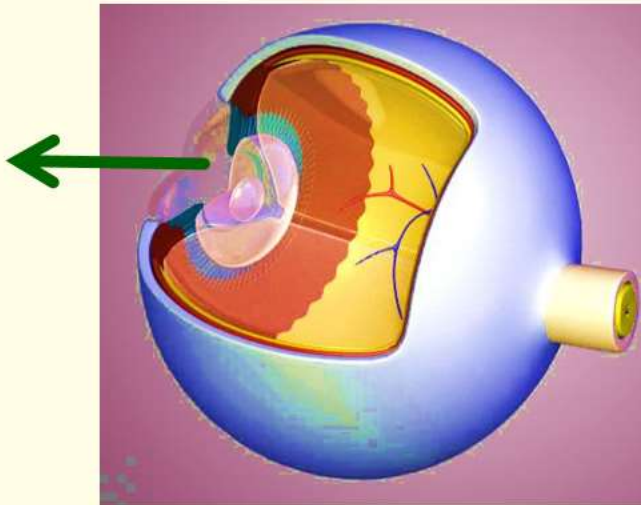
- In anatomy, the orbit is the cavity or socket of the skull in which the eye and its appendages are situated.
- "Orbit" can refer to the bony socket, or it can also be used to imply the contents.
- In the adult human, the volume of the orbit is 30 mL, of which the eye occupies 6.5 mL.



THE OUTER LAYERS OF THE EYE

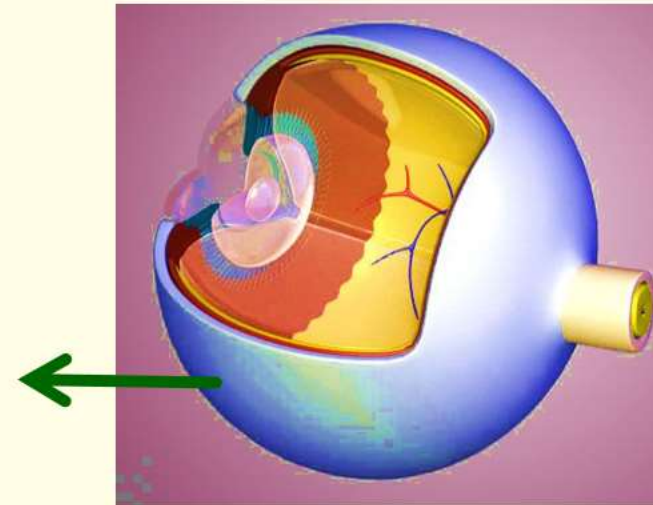
CORNEA

The transparent front part of the eye that covers iris, pupil and the anterior chamber.

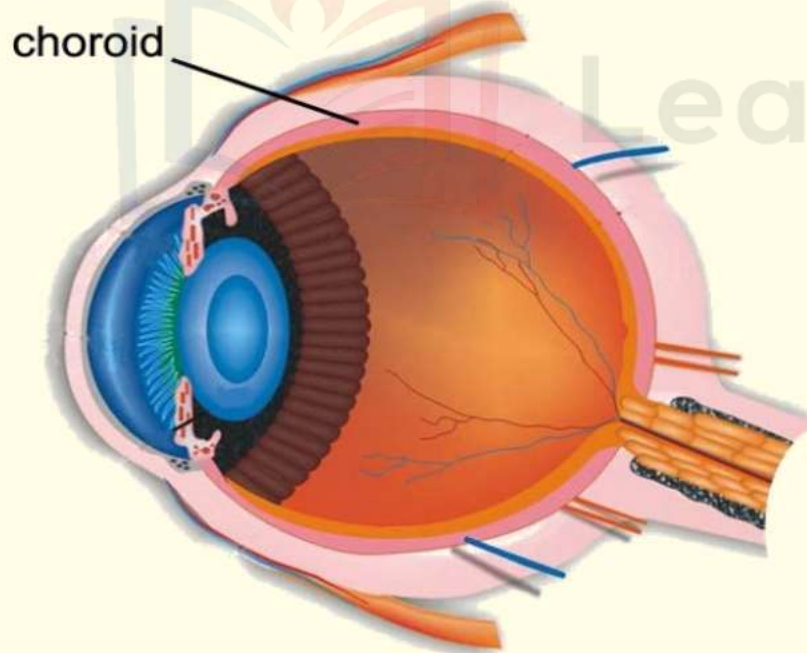


CONJUCTIVA

The membrane that lines the eyelids and covers the eyeball except the cornea.



THE OUTER LAYERS OF THE EYE



SCLERA

The white outer layer of the eyeball. At the front of the eye it is continuous with the cornea.

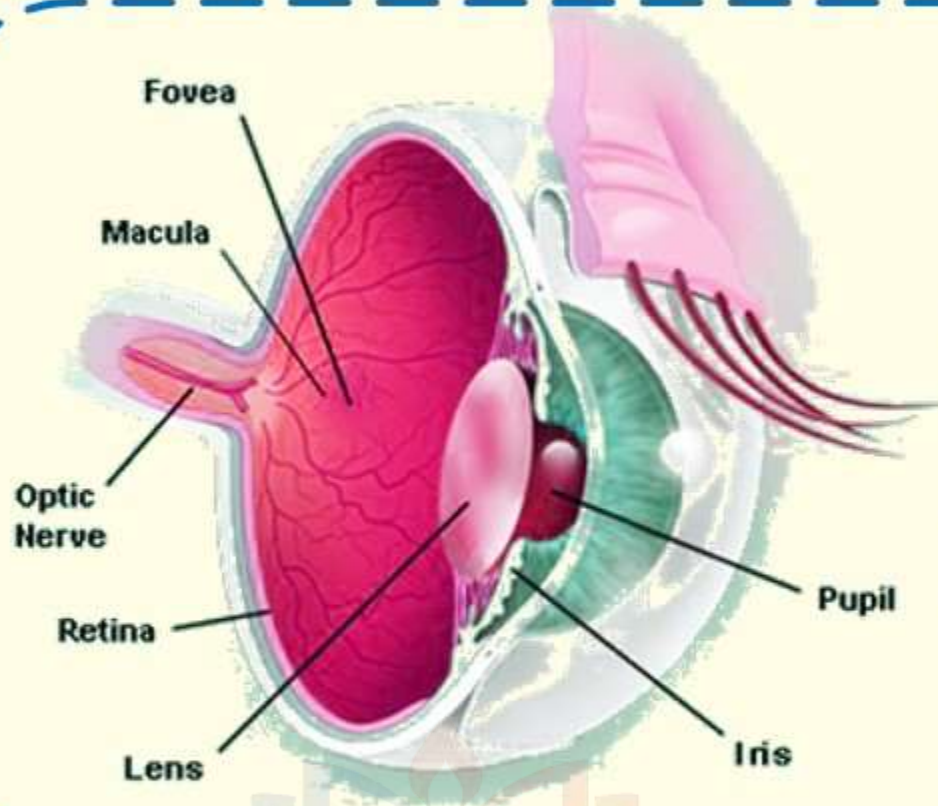
CHOROID

The choroid is the second layer after sclera containing connective tissue. It is the layer prior to retina





THE LIGHT SENSITIVE INNER COAT - RETINA

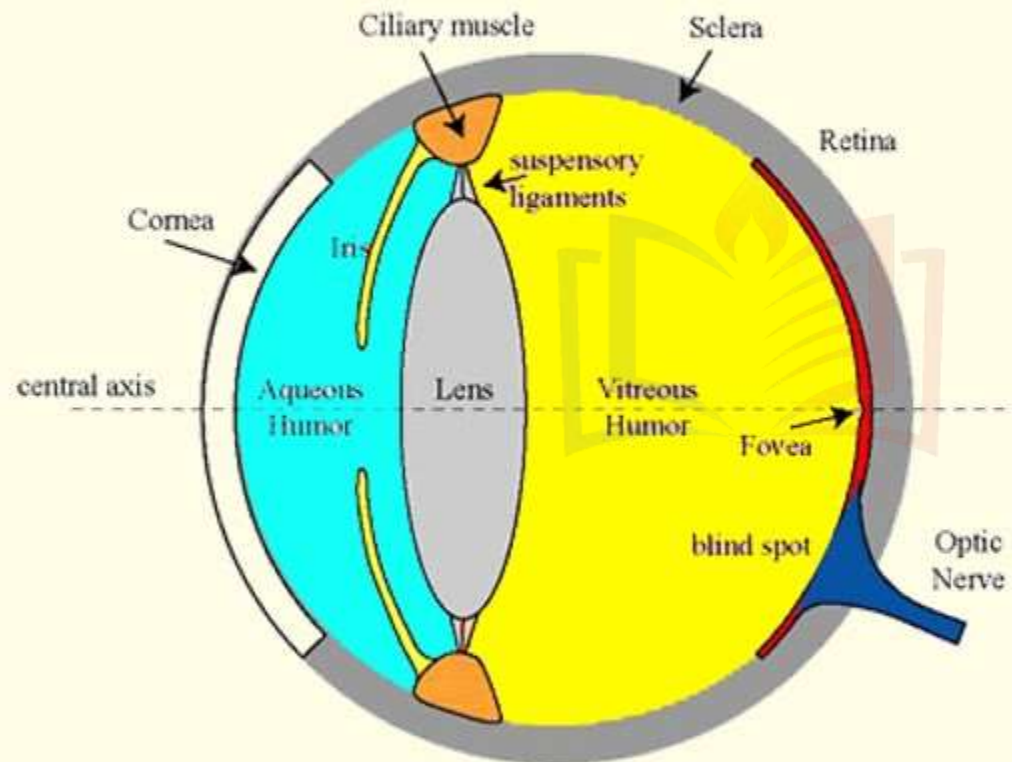


- Retina is a light sensitive screen inside the eye on which image is formed.
- It contains 2 photoreceptor cells
 - Rod cells and
 - Cone cells.
- Rod cells help to see in dim light.
- Cone cells are responsible for colour vision and helps to see in bright light.

Macula surrounding the Fovea is the area of keenest visual activity concentrated with cone cells. The point of entry of the optic nerves on the retina, insensitive to light is called the blind spot.



THE INNER FLUIDS OF THE EYE BALL



AQUEOUS HUMOUR

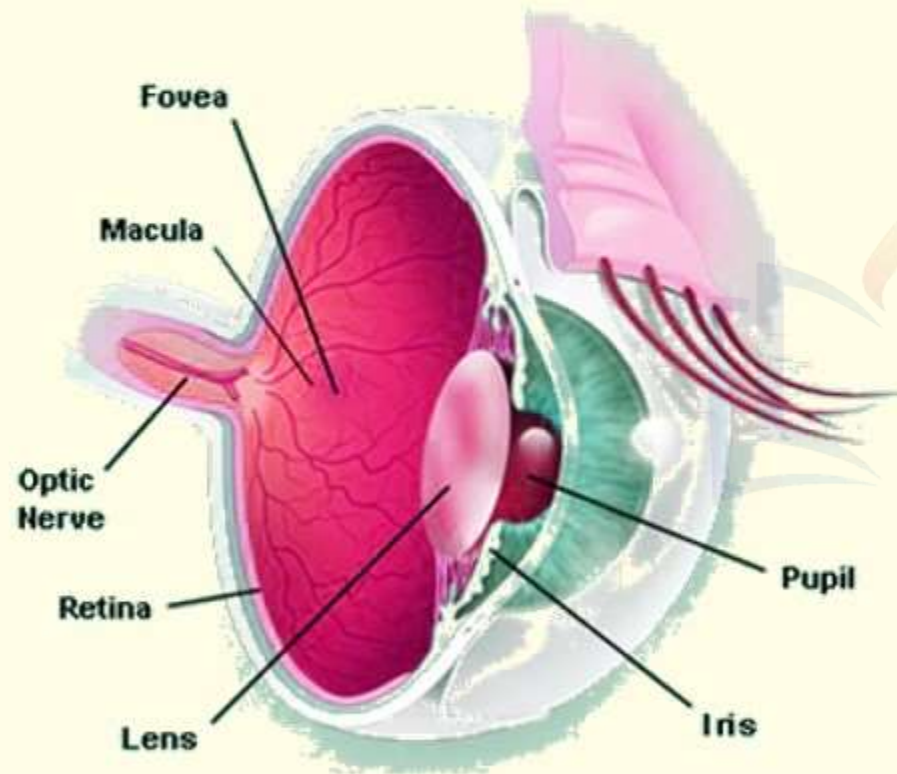
It is the fluid that fills the space between cornea and eye lens.

VITREOUS HUMOUR

It is the fluid that fills the space between the lens and the retina of the eyeball.



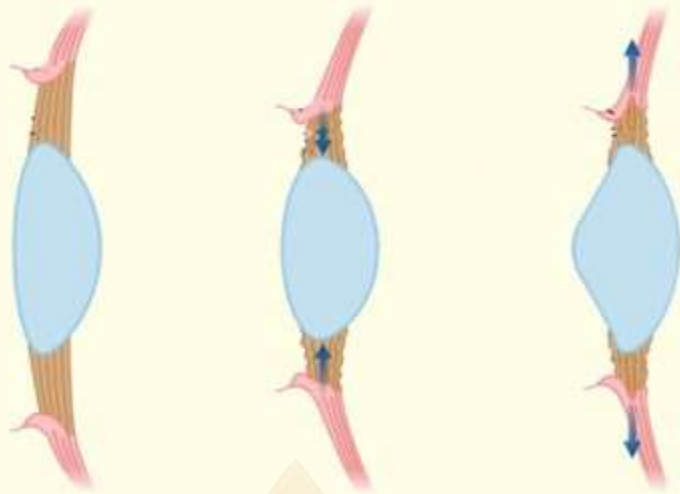
APERTURE OF THE HUMAN EYE – THE EYE LENS



- 1** The eye lens is a **Convex lens** made up of transparent and flexible jelly like material. Its curvature can be adjusted with the help of ciliary muscles.
- 2** **Ciliary muscles** are attached to the eye lens which can modify the shape of the eye lens.
- 3** **Pupil** is the hole in the middle of the iris membrane through which light enters the eye. It appears black because light falling on it goes into the eye and doesn't come back.



THE HUMAN EYE - POWER OF ACCOMMODATION



Eye lens with the accommodating feature.

- Accommodation power is the ability of eye to change the focal length of the eye lens with the help of ciliary muscles.
- The near point of the eye is 25 cm. this is the nearest point of clear vision for a normal healthy eye.
- The far point of the eye is at infinity.

More bulge of the lens means a shorter focal length and thinner lens means a longer focal length.



CONDITIONS THAT OCCUR DUE TO IMPROPER ACCOMODATION OF THE EYE

MYOPIA



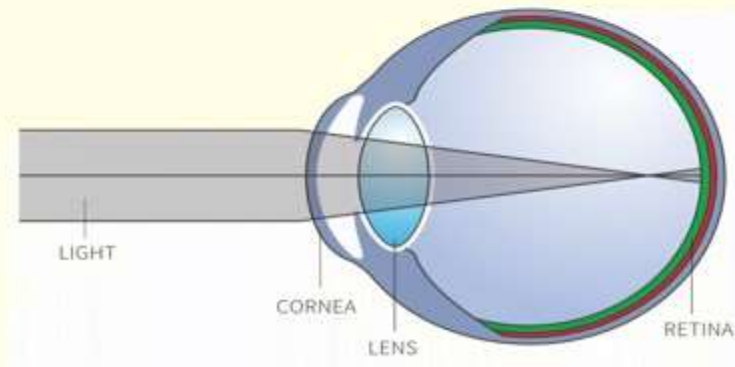
HYPERMETROPIA





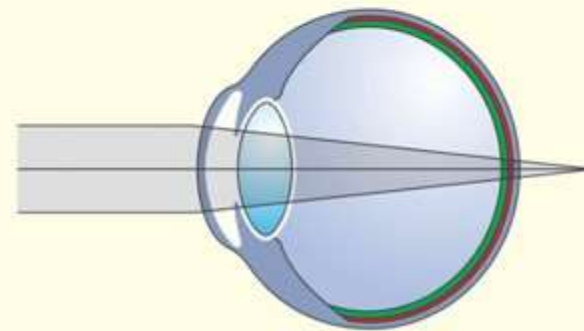
MYOPIA

- The image is formed in front of the retina.
- Myopia is due to :
 - Excessive curvature of eye lens
 - Elongation of the eye ball
 - Thus, a person can see near objects clearly only



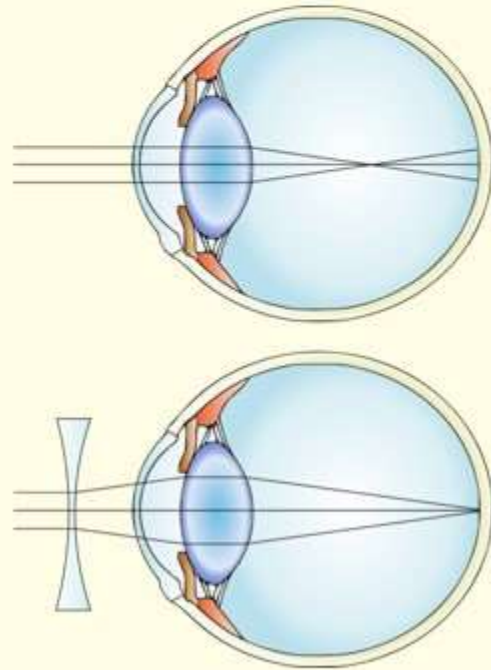
HYPERMETROPIA

- The image is formed at the back of the retina
- Hypermetropia is due to :
 - Focal length of eye lens increases
 - Shortening of the eye ball
 - Thus, a person can see distant object clearly

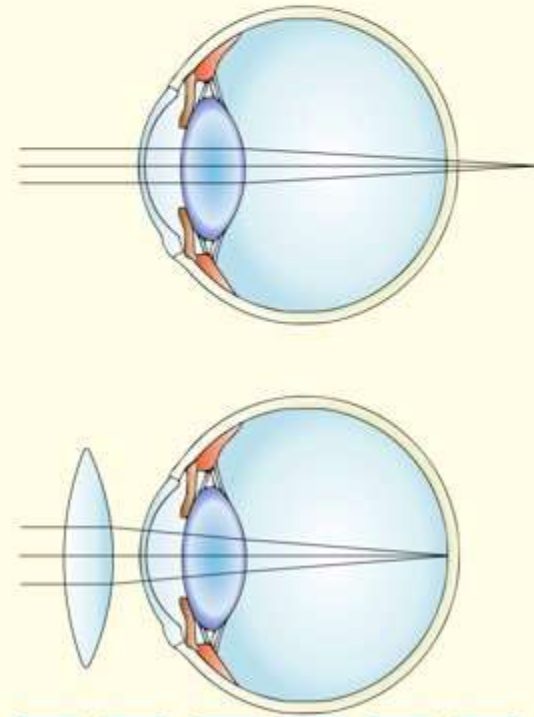


**CORRECTION OF THE CONDITIONS****MYOPIA**

Corrected by a diverging lens
Concave lens with minus power

**HYPERMETROPIA**

Corrected by a converging lens
Convex lens with plus power





OTHER CONDITIONS OF THE EYE

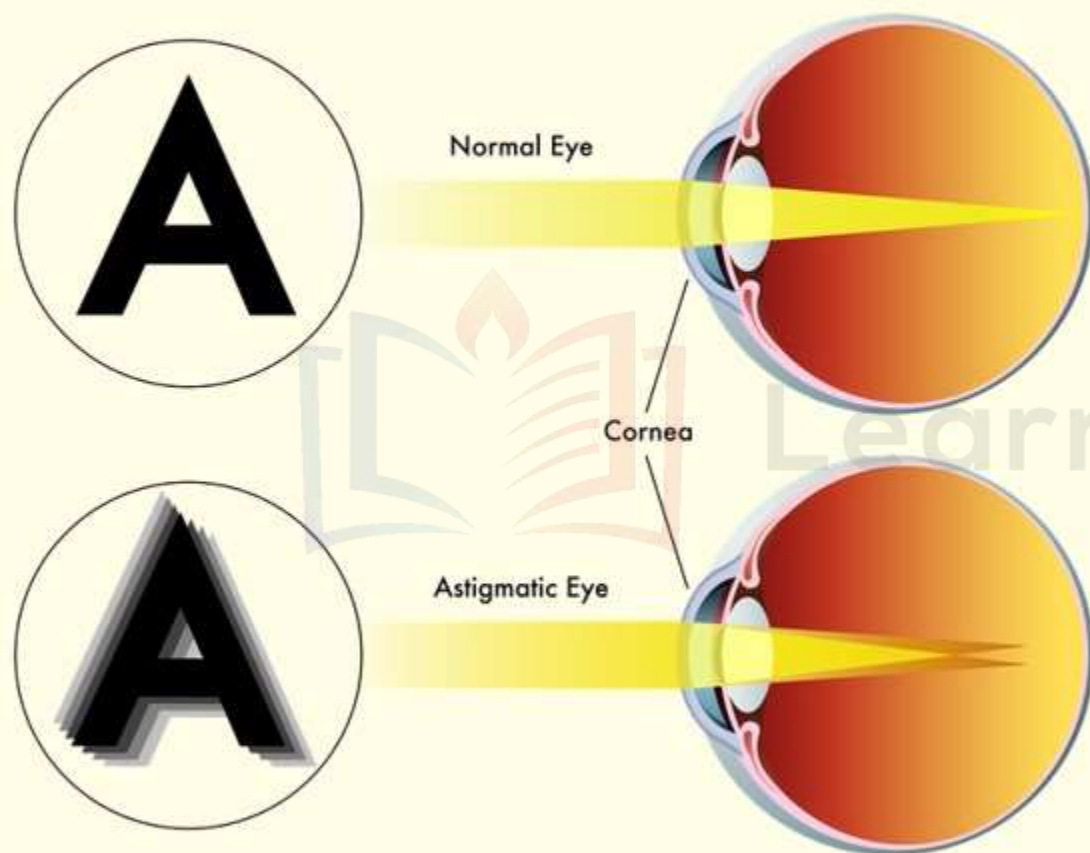
PRESBIOPIA



- It is a kind of defect in human eye which occurs due to ageing.
- It happens due to
 - Decrease in flexibility of eye lens
 - Gradual weakening of the ciliary muscles.
- Presbiopia cannot be cured but the condition can be reversed by using
 - Bifocal lens
 - Contact lens'
 - Surgery and inserting intraocular lens



ASTIGMATISM



- Astigmatism is a kind of defect in human eye due to which a person cannot see or focus horizontal and vertical lines simultaneously.
- It occurs due to irregular curvature of the lens and is sometimes genetic.
- Laser eye surgery (LASIK and PRK) is successful in treating astigmatism.

CATARACT



- Due to membrane growth over eye lens, the eye lens becomes hazy and opaque.
- This leads to partial or complete loss of vision.
- Cataract is only corrected by surgery.



ASPHERIC LENS AND THEIR USAGE



Distortion Free Image



Distorted Image



- Aspheric lens surface can reduce or eliminate distortion in an image or spherical aberration and optical aberrations(astigmatism).
- A single aspheric lens can often replace a much more complex multi-lens system. The resulting device is smaller and lighter, and sometimes cheaper than the multi-lens
- Moreover, the reduction of the magnification effect of a lens helps patients with different powers in the 2 eyes (anisometropia).
- They will also not distort the eye shape of the patient thus giving a better aesthetic appearance.
- Aspheric lens is also used in zoom lens, astronomical telescope, high power laser, LED lens, high-quality magnifier etc.





COLOUR BLINDNESS

Normal Eye



RED



GREEN



YELLOW

Colorblind Eye



RED



GREEN



YELLOW

- Colour blindness is the condition of decreased ability to see colour or differentiate colors.
- The most common cause of color blindness is an inherited fault in the development of one or more of the three sets of **color sensing cones in the eye**.
- Males are more likely to be color blind than females.



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